

# Practical # 01

## Objective:

Construct and Implement C++ classes and templates using Code::Blocks IDE.

## Theory / Conceptual Review:

To provide a quick refresher on C++ programming concepts, a **Rectangle Class** is constructed and implemented with class declaration, member function definition, and function calling in the main program.

### Rectangle Class Pseudo Code:

- ✓ Declare a class Rect, with data members: length, breadth, area, and perimeter.
- ✓ Declare member functions to accept input and compute area and perimeters: input(), ComputeArea(), ComputePeri(), Display()
- ✓ Write a main function:
  - instantiate the class by passing appropriate arguments to the constructor
  - invoke ComputeArea(), ComputePeri() and Display() functions

### Program in C++:

```
#include<iostream>
using namespace std;
class Rect{
public:
    int length, breadth, area, perimeter;

    void input() {
        cout << "Enter length of rectangle:";
        cin >> length;
        cout << "Enter breadth of rectangle:";
        cin >> breadth;
    }

    void ComputeArea() {
        area = length * breadth;
    }

    void ComputePerimeter() {
        perimeter = 2 * (length + breadth) ;
    }

    void display() {
        cout << "Area of rectangle is:" << area;
        cout << "\nPerimeter of rectangle is:" << perimeter;
    }
};

int main() {
    Rect obj;

    obj.input();
    obj.ComputeArea();
    obj.ComputePerimeter();
    obj.display();

    return 0;
}
```

### OUTPUT:

```
Enter length of rectangle:4
Enter breadth of rectangle:8
Area of rectangle is:32
Perimeter of rectangle is:24
```

**Review Questions/ Exercise:**

1. **Develop** a class Student with attributes (*name, roll number, marks in three subjects*). Add member functions to calculate total marks and percentage, and display the result.
2. **Implement** a class BankAccount with attributes (*account number, holder name, balance*) and member functions deposit(), withdraw(), and displayBalance().
3. **Construct** a class Temperature with member functions to convert Celsius to Fahrenheit and Fahrenheit to Celsius.
4. **Develop** a class Calculator that uses **function overloading** to perform addition, subtraction, multiplication, and division.
5. **Implement** a class Complex to perform arithmetic operations (addition, subtraction, multiplication) on complex numbers using member functions.

**Name:** \_\_\_\_\_

**Roll #:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Subject Teacher**